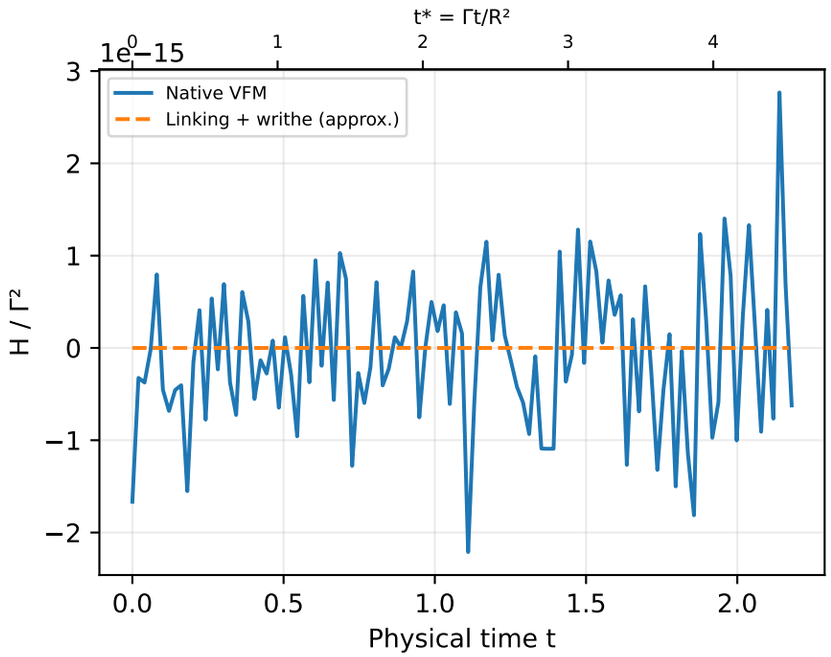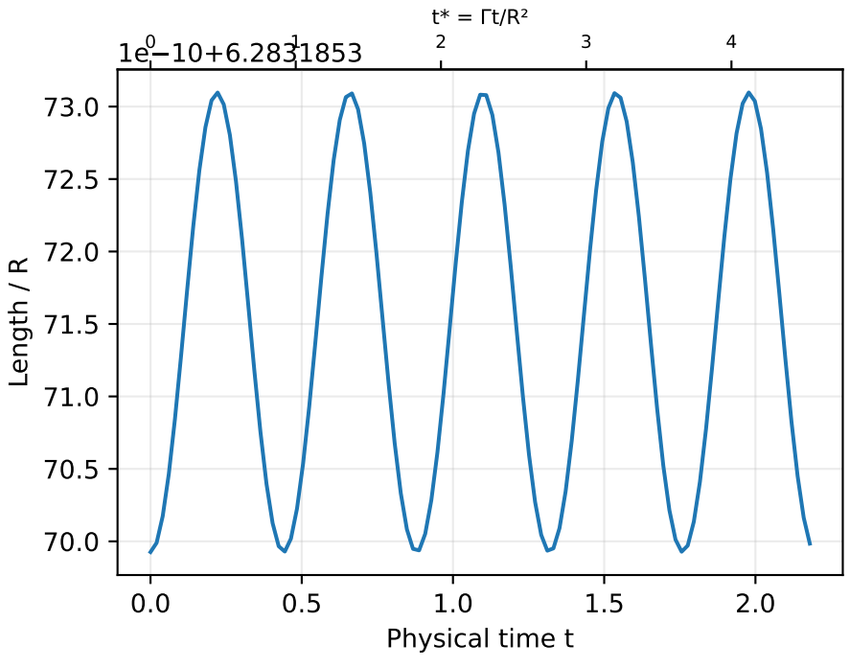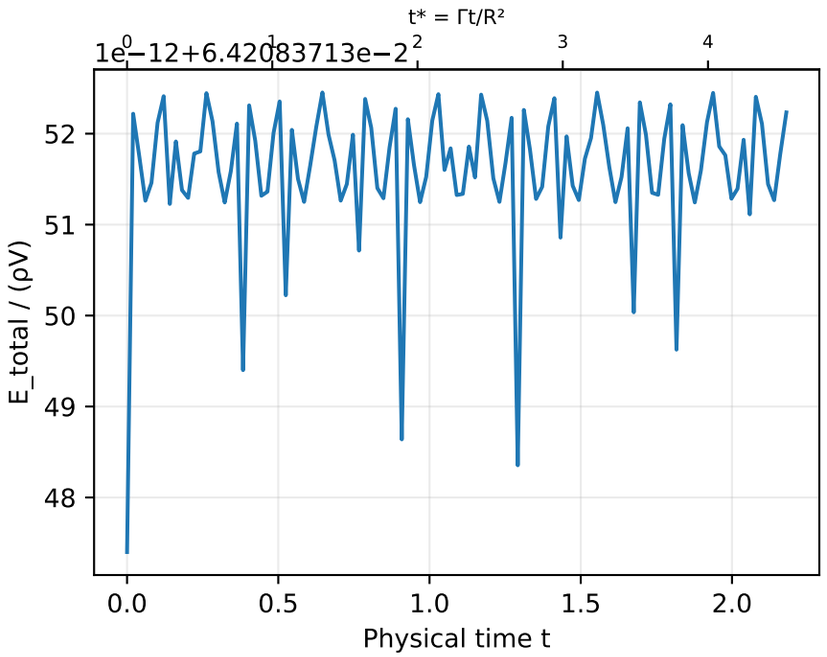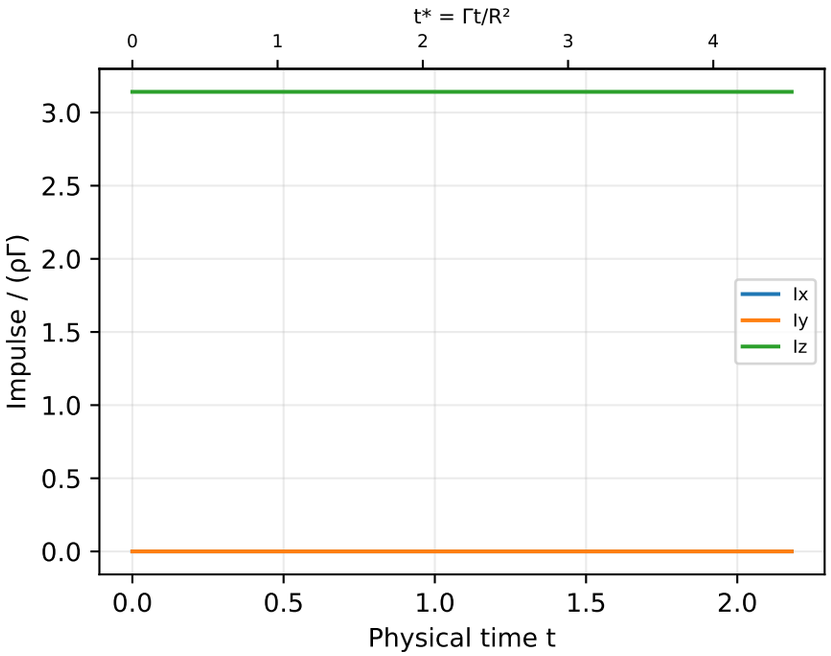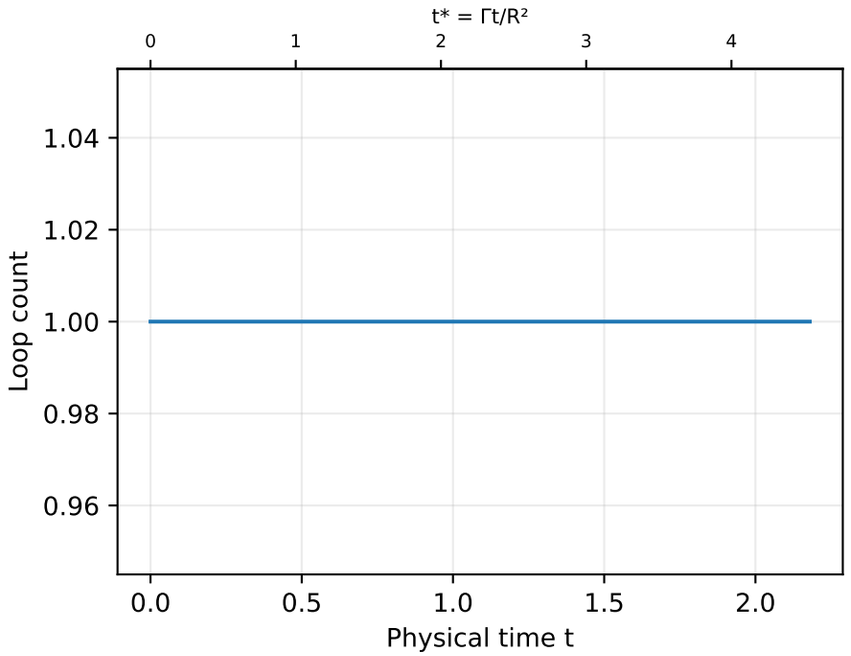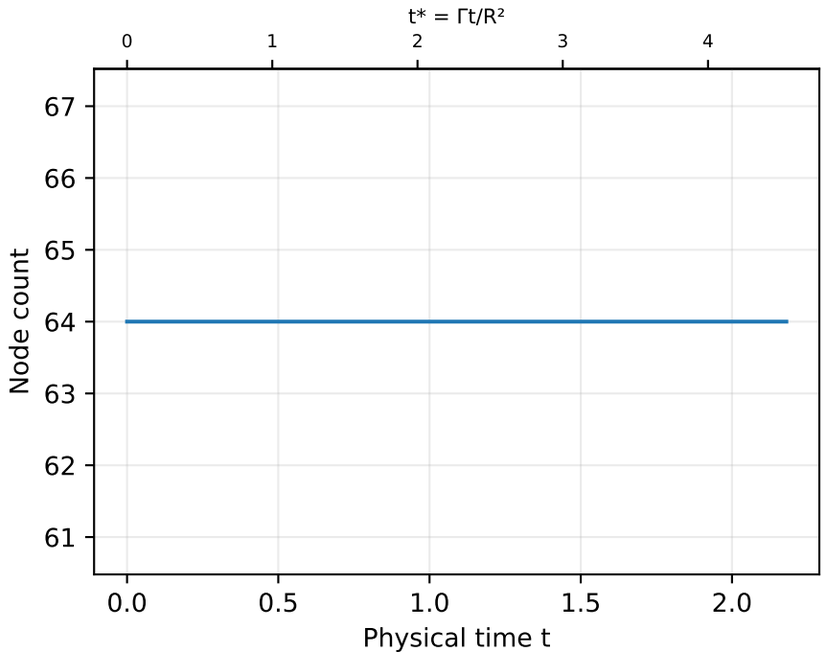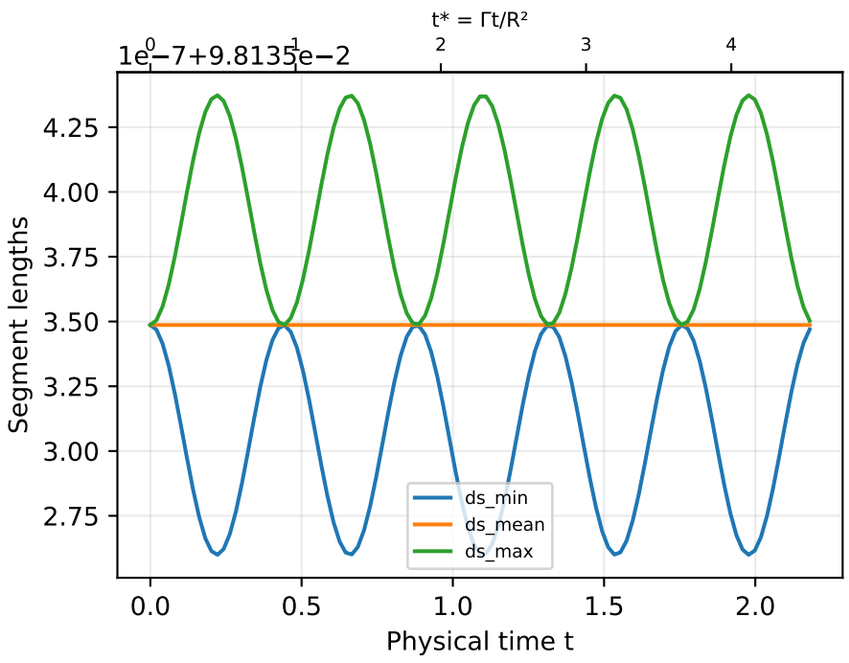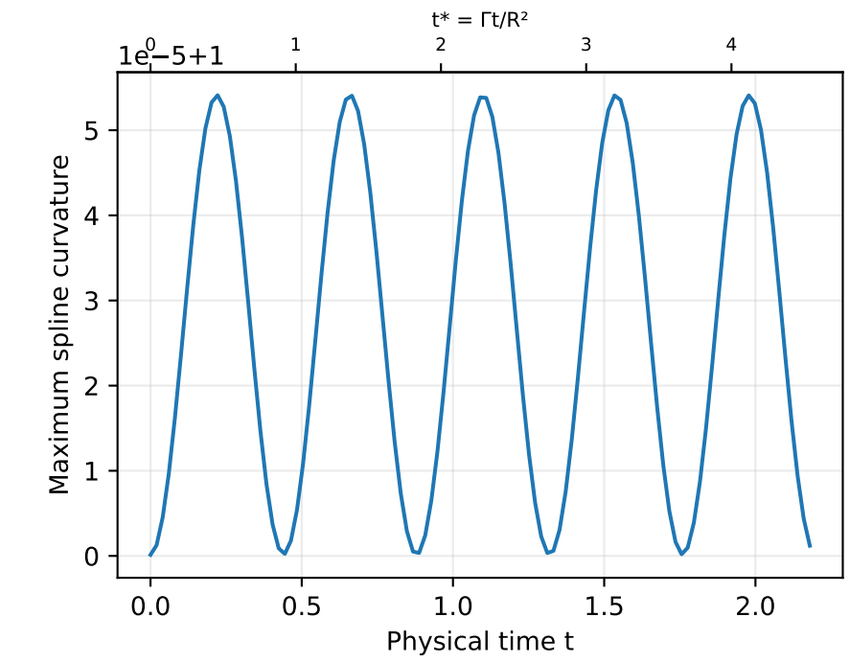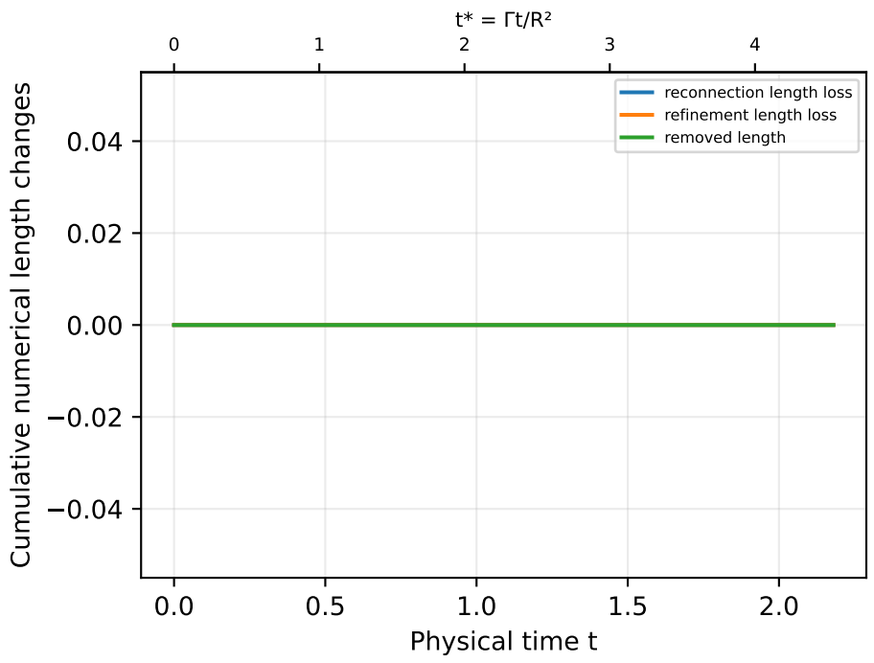single_ring · periodic Biot–Savart 64³ mesh · a=0.001 · no mutual friction